

UD65640

Application Note

Register

AN-K0001-251230

|                                                                      |           |
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# 1. Target

Introducing the usage of UD65640 Register and OTP.

## 2. Manufacture Command

### 2.1 Command list

|                        | Manufacture Command List |                  |             |                  |            |             |                   |             |              |         |
|------------------------|--------------------------|------------------|-------------|------------------|------------|-------------|-------------------|-------------|--------------|---------|
| Command                | W / R                    | B7               | B6          | B5               | B4         | B3          | B2                | B1          | B0           | default |
| MCS Ctrl(80h)          | W / R                    | 0                | 0           | 0                | 0          | 0           | 0                 | 0           | MCS_EN       | 00h     |
| CHIP_ID(81h)           | RO                       | CHIPID[07:00]    |             |                  |            |             |                   |             |              | 65h     |
|                        | RO                       | CHIPID[15:08]    |             |                  |            |             |                   |             |              | 64h     |
|                        | RO                       | CHIPID[23:16]    |             |                  |            |             |                   |             |              | 0Ah     |
|                        | RO                       | CHIPID[31:24]    |             |                  |            |             |                   |             |              | 01h     |
| OTP Ctrl(82h)          | W / R                    | GRP8             | GRP7        | GRP6             | GRP5       | GRP4        | GRP3              | GRP2        | GRP1         | 00h     |
|                        | W / R                    | X                | GRP15       | GRP14            | GRP13      | GRP12       | GRP11             | GRP10       | GRP9         | 00h     |
|                        | W / R                    | X                | X           | X                | X          | GRP19       | GRP18             | GRP17       | GRP16        | 00h     |
|                        | W / R                    | 0                | 0           | 0                | OTPEN      | 0           | OTP_ACT_MODE[2:0] |             |              | 00h     |
| OTP State(83h)         | RO                       | 0                | 0           | 0                | 0          | 0           | 0                 | 0           | 0            | 00h     |
|                        | RO                       | 0                | 0           | G19_WC           | G15_WC     | WRALLFIN    | 0                 | 0           | LOADALLFIN   | 00h     |
|                        | RO                       | G4_WC[2:0]       |             |                  | G3_WC[1:0] |             | G2_WC[1:0]        |             | G1_WC        | 00h     |
|                        | RO                       | G14_WC           | G7_WC[2:0]  |                  |            | G6_WC[1:0]  |                   | G5_WC[1:0]  |              | 00h     |
|                        | RO                       | G13_WC           | G12_WC[1:0] |                  | G11_WC     | G9_WC[1:0]  |                   | G8_WC[1:0]  |              | 00h     |
|                        | RO                       | 0                | 0           | G10_WC[5:0]      |            |             |                   |             |              | 00h     |
| OTP Power(85h)         | W / R                    | 0                | 0           | 0                | 0          | 0           | 0                 | REF_OTP_EN  | VREFP_OTP_EN | 00h     |
| OTP State 1(87h)       | RO                       | 0                | 0           | G18_WC[1:0]      |            | G17_WC[1:0] |                   | G16_WC[1:0] |              | 00h     |
| Device Code(88h)       | W / R                    | DDB0[7:0]        |             |                  |            |             |                   |             |              | 00h     |
|                        | W / R                    | DDB1[7:0]        |             |                  |            |             |                   |             |              | 00h     |
|                        | W / R                    | DDB2[7:0]        |             |                  |            |             |                   |             |              | 00h     |
| Interface Control(8Dh) | W / R                    | 1                | 0           | DM               | 0          | 0           | 0                 | 0           | 0            | 80h     |
| AVEE Control(96h)      | W / R                    | DV_AVEE_OPT[3:0] |             |                  |            | 0           | 0                 | 0           | AVEE_ON      | 31h     |
| VEMR Control(97h)      | W / R                    | DV_VEMR_OPT[3:0] |             |                  |            | 0           | 0                 | 0           | VEMR_ON      | 01h     |
| SWIRE Control(98h)     | W / R                    | 0                | 0           | SWIRE_CKSEL[1:0] | PS2_EN     | PS1_EN      | SWIRE_OPT         | SWIRE_ON    |              | 07h     |
|                        | W / R                    | PS1_NUM[7:0]     |             |                  |            |             |                   |             |              | 05h     |
|                        | W / R                    | PS2_NUM[7:0]     |             |                  |            |             |                   |             |              | 4Fh     |
|                        | W / R                    | SFT_DLY[7:0]     |             |                  |            |             |                   |             |              | 04h     |

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|                                      |       |                     |    |           |           |                 |           |   |        |     |
|--------------------------------------|-------|---------------------|----|-----------|-----------|-----------------|-----------|---|--------|-----|
|                                      | W / R | PS_H_WIDTH[3:0]     |    |           |           | PS_L_WIDTH[3:0] |           |   |        | 22h |
| Image Engine Control(A0h)            | W / R | 0                   | 0  | 0         | DBV_M ODE | 0               | 0         | 0 | BC_OFF | 00h |
|                                      | W / R | INIT_DBV_DSPON[7:0] |    |           |           |                 |           |   |        | 80h |
| Panel Driving Settings1(A1h)         | W / R | RTN[7:0]            |    |           |           |                 |           |   |        | 64h |
|                                      | W / R | 0                   | 0  | 0         | 0         | 0               | RTN[10:8] |   |        | 00h |
|                                      | W / R | VBP[7:0]            |    |           |           |                 |           |   |        | 08h |
|                                      | W / R | VFP[7:0]            |    |           |           |                 |           |   |        | 0Ch |
|                                      | W / R | HBP[7:0]            |    |           |           |                 |           |   |        | 0Ch |
|                                      | W / R | HFP[7:0]            |    |           |           |                 |           |   |        | 16h |
| Panel Driving Settings2(A2h)         | W / R | FV                  | FH | CGM[1:0]  |           |                 | SPM[2:0]  |   |        | 00h |
| Pixel control data infor(A8h)        | W / R | INDEX[7:0]          |    |           |           |                 |           |   |        | 00h |
| Pixel control byte write (A9h)       | W / R | 0                   | 0  | 0         | 0         | 0               | 0         | 0 | WE     | 00h |
| Pixel control data information (AAh) | W / R | 0                   | 0  | TLSB[5:0] |           |                 |           |   |        | 04h |

## 2.2 Command

### MCS Ctrl: Manufacturer Command Set Control (80h)

| 80h          | MCS Ctrl                                                                                                                                                                                                    |    |    |    |    |    |    |    |        |     |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|--------|-----|
|              | W / R                                                                                                                                                                                                       | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0     | Hex |
| Command      | —                                                                                                                                                                                                           | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 80h |
| parameter 1  | W / R                                                                                                                                                                                                       | 0  | 0  | 0  | 0  | 0  | 0  | 0  | MCS_EN | 00h |
| Descriptions | MCS_EN<br>This command releases parameters so that Manufacturer Command inputs are enabled.<br>1: Access enabled.<br>0: Access denied. (Protection on)<br>1 = Must fill 1; 0 = Must fill 0; X = Don't care. |    |    |    |    |    |    |    |        |     |
| Restrictions |                                                                                                                                                                                                             |    |    |    |    |    |    |    |        |     |

### CHIP\_ID: Chip ID Read (81h)

| 81h          | CHIP ID                                                                                                                        |    |    |    |    |    |    |    |    |     |
|--------------|--------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|              | RO                                                                                                                             | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 | Hex |
| Command      | —                                                                                                                              | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 81h |
| parameter 1  | RO                                                                                                                             | 0  | 1  | 1  | 0  | 0  | 1  | 0  | 1  | 65h |
| parameter 2  | RO                                                                                                                             | 0  | 1  | 1  | 0  | 0  | 1  | 0  | 0  | 64h |
| parameter 3  | RO                                                                                                                             | 0  | 0  | 0  | 0  | 1  | 1  | 0  | 0  | 0Ah |
| parameter 4  | RO                                                                                                                             | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 01h |
| Descriptions | This command is used to read the chip ID of current revision. (Read Only)<br>1 = Must fill 1; 0 = Must fill 0; X = Don't care. |    |    |    |    |    |    |    |    |     |
| Restrictions |                                                                                                                                |    |    |    |    |    |    |    |    |     |

## OTP Ctrl: OTP Control (82h)

| 82h         | OTP Ctrl                                                                                              |      |               |       |       |       |                   |       |       |     |
|-------------|-------------------------------------------------------------------------------------------------------|------|---------------|-------|-------|-------|-------------------|-------|-------|-----|
|             | W / R                                                                                                 | B7   | B6            | B5    | B4    | B3    | B2                | B1    | B0    | Hex |
| Command     | —                                                                                                     | 1    | 0             | 0     | 0     | 0     | 0                 | 1     | 0     | 82h |
| parameter 1 | W / R                                                                                                 | GRP8 | GRP7          | GRP6  | GRP5  | GRP4  | GRP3              | GRP2  | GRP1  | 00h |
| parameter 2 | W / R                                                                                                 | 0    | GRP15         | GRP14 | GRP13 | GRP12 | GRP11             | GRP10 | GRP9  | 00h |
| parameter 3 | W / R                                                                                                 | 0    | 0             | 0     | 0     | GRP19 | GRP18             | GRP17 | GRP16 | 00h |
| parameter 4 | W / R                                                                                                 | 0    | 0             | 0     | OTPEN | 0     | OTP_ACT_MODE[2:0] |       |       | 00h |
| Description | OTP_ACT_MODE                                                                                          |      |               |       |       |       |                   |       |       |     |
|             | Operation mode for OTP. UD65640 only supports program OTP in sleep-in mode.                           |      |               |       |       |       |                   |       |       |     |
|             |                                                                                                       |      |               |       |       |       |                   |       |       |     |
|             | OTP_ACT_MODE[2:0]                                                                                     |      | Function      |       |       |       |                   |       |       |     |
|             | 3'b000                                                                                                |      | Program group |       |       |       |                   |       |       |     |
| Description |                                                                                                       |      |               |       |       |       |                   |       |       |     |
|             | OTPEN                                                                                                 |      |               |       |       |       |                   |       |       |     |
|             | Enable for OTP access, please be sure to set OTP control registers and data before enable OTP access. |      |               |       |       |       |                   |       |       |     |
|             | 1: Enabled.                                                                                           |      |               |       |       |       |                   |       |       |     |
|             | 0: Disabled.                                                                                          |      |               |       |       |       |                   |       |       |     |
| Description |                                                                                                       |      |               |       |       |       |                   |       |       |     |
|             | GRP1 ~ GRP19: Enable requires program the group.                                                      |      |               |       |       |       |                   |       |       |     |
|             | 1: Enabled.                                                                                           |      |               |       |       |       |                   |       |       |     |
|             | 0: Disabled.                                                                                          |      |               |       |       |       |                   |       |       |     |
|             |                                                                                                       |      |               |       |       |       |                   |       |       |     |
| Description |                                                                                                       |      |               |       |       |       |                   |       |       |     |
|             | 1 = Must fill 1; 0 = Must fill 0; X = Don't care.                                                     |      |               |       |       |       |                   |       |       |     |
|             |                                                                                                       |      |               |       |       |       |                   |       |       |     |
|             |                                                                                                       |      |               |       |       |       |                   |       |       |     |
|             |                                                                                                       |      |               |       |       |       |                   |       |       |     |
| Restriction | Note:                                                                                                 |      |               |       |       |       |                   |       |       |     |

## OTP State: OTP State (83h)

| 84h         | OTP State                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |             |             |            |            |            |            |            |     |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|-------------|------------|------------|------------|------------|------------|-----|
|             | RO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | B7         | B6          | B5          | B4         | B3         | B2         | B1         | B0         | Hex |
| Command     | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1          | 0           | 0           | 0          | 0          | 0          | 1          | 1          | 83h |
| parameter 1 | RO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0          |             |             |            |            |            |            |            | 00h |
| parameter 2 | RO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0          | 0           | G19_WC      | G15_WC     | WRALLFIN   | 0          | 0          | LOADALLFIN | 00h |
| parameter 3 | RO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | G4_WC[2:0] |             |             | G3_WC[1:0] |            | G2_WC[1:0] |            | G1_WC      | 00h |
| parameter 4 | RO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | G14_WC     | G7_WC[2:0]  |             |            | G6_WC[1:0] |            | G5_WC[1:0] |            | 00h |
| parameter 5 | RO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | G13_WC     | G12_WC[1:0] |             | G11_WC     | G9_WC[1:0] |            | G8_WC[1:0] |            | 00h |
| parameter 6 | RO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0          | 0           | G10_WC[5:0] |            |            |            |            |            | 00h |
| Description | <div>LOADALLFIN<br/>Load-all-data from OTP finish indicator. It will clear after read.</div> <div>WRALLFIN<br/>Write-all operation finish indicator. It will clear after read.</div> <div>G1_WC[1:0]<br/>Group 1 read the number of burn-ins, max is 1.</div> <div>G2_WC[1:0]<br/>Group 2 read the number of burn-ins, max is 2.</div> <div>G3_WC[1:0]<br/>Group 3 read the number of burn-ins, max is 2.</div> <div>G4_WC[1:0]<br/>Group 4 read the number of burn-ins, max is 3.</div> <div>G5_WC[1:0]<br/>Group 5 read the number of burn-ins, max is 2.</div> <div>G6_WC[5:0]<br/>Group 6 read the number of burn-ins, max is 2.</div> <div>G7_WC[2:0]<br/>Group 7 read the number of burn-ins, max is 3.</div> <div>G10_WC[5:0]<br/>Group 10 read the number of burn-ins, max is 6.</div> <div>G11_WC<br/>Group 11 read the number of burn-ins, max is 1.</div> <div>G12_WC[1:0]<br/>Group 12 read the number of burn-ins, max is 2.</div> <div>G13_WC<br/>Group 13 read the number of burn-ins, max is 1.</div> <div>1 = Must fill 1; 0 = Must fill 0; X = Don't care.</div> |            |             |             |            |            |            |            |            |     |

## OTP Power: OTP Power (85h)

| 85h         | OTP Power                                                                                                                                                                                                                                                                                                                                                                                                                        |    |    |    |    |    |    |            |              |     |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|------------|--------------|-----|
|             | W / R                                                                                                                                                                                                                                                                                                                                                                                                                            | B7 | B6 | B5 | B4 | B3 | B2 | B1         | B0           | Hex |
| Command     | —                                                                                                                                                                                                                                                                                                                                                                                                                                | 1  | 0  | 0  | 0  | 0  | 1  | 0          | 1            | 85h |
| parameter 1 | W / R                                                                                                                                                                                                                                                                                                                                                                                                                            | 0  | 0  | 0  | 0  | 0  | 0  | REF_OTP_EN | VREFP_OTP_EN | 00h |
| Description | <p>REF_OTP_EN: Enable OTP reference voltage before program OTP.<br/>1: Enabled.<br/>0: Disabled.</p> <p>VREFP_OTP_EN:<br/>1: Enabled OTP VREFP voltage before program OTP.<br/>0: Disabled OTP VREFP voltage before program OTP.</p> <p>Note: UD65640 only supports program OTP in sleep-in mode, so REF_OTP_EN and VREFP_OTP_EN needs to be started in sleep mode.</p> <p>1 = Must fill 1; 0 = Must fill 0; X = Don't care.</p> |    |    |    |    |    |    |            |              |     |
| Restriction |                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |    |    |    |    |    |            |              |     |

## OTP State 1: OTP State 1 (87h)

| 87h         | OTP State 1                                                                                                                                                                                                                      |    |    |             |    |             |    |             |    |     |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-------------|----|-------------|----|-------------|----|-----|
|             | RO                                                                                                                                                                                                                               | B7 | B6 | B5          | B4 | B3          | B2 | B1          | B0 | Hex |
| Command     | —                                                                                                                                                                                                                                | 1  | 0  | 0           | 0  | 0           | 1  | 1           | 1  | 87h |
| parameter 1 | RO                                                                                                                                                                                                                               | 0  | 0  | G18_WC[1:0] |    | G17_WC[1:0] |    | G16_WC[1:0] |    | 00h |
| Description | <div>G16_WC[1:0]<br/>Group 16 read the number of burn-ins, max is 2.</div> <div>G17_WC[1:0]<br/>Group 17 read the number of burn-ins, max is 2.</div> <div>G18_WC[1:0]<br/>Group 18 read the number of burn-ins, max is 2.</div> |    |    |             |    |             |    |             |    |     |
| Restriction |                                                                                                                                                                                                                                  |    |    |             |    |             |    |             |    |     |



**Device Code: Device Code(88h)**

| 88h         | Device Code                                                                                                                                                                                                                                                                   |           |    |    |    |    |    |    |    |     |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----|----|----|----|----|----|----|-----|
|             | W / R                                                                                                                                                                                                                                                                         | B7        | B6 | B5 | B4 | B3 | B2 | B1 | B0 | Hex |
| Command     | —                                                                                                                                                                                                                                                                             | 1         | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 88h |
| parameter 1 | W / R                                                                                                                                                                                                                                                                         | DDB0[7:0] |    |    |    |    |    |    |    | 00h |
| parameter 2 | W / R                                                                                                                                                                                                                                                                         | DDB1[7:0] |    |    |    |    |    |    |    | 00h |
| parameter 3 | W / R                                                                                                                                                                                                                                                                         | DDB2[7:0] |    |    |    |    |    |    |    | 00h |
| Description | <p>DDB0:<br/>This command is for user command(DAh) version number.</p> <p>DDB1:<br/>This command is for user command(DBh) version number.</p> <p>DDB2:<br/>This command is for user command(DCh) version number.</p> <p>1 = Must fill 1; 0 = Must fill 0; X = Don't care.</p> |           |    |    |    |    |    |    |    |     |
| Restriction |                                                                                                                                                                                                                                                                               |           |    |    |    |    |    |    |    |     |

**Interface Control: Interface Control(8Dh)**

| 8Dh         | Interface Control                                                                                                                          |    |    |    |    |    |    |    |    |     |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|             | W / R                                                                                                                                      | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 | Hex |
| Command     | —                                                                                                                                          | 1  | 0  | 0  | 0  | 1  | 1  | 0  | 1  | 8Dh |
| parameter 1 | W / R                                                                                                                                      | 1  | 0  | DM | 0  | 0  |    | 0  |    | 80h |
| parameter 2 | W / R                                                                                                                                      | 0  | 0  | 0  | 0  | 0  | 0  |    |    | 00h |
| Description | DM:<br>0: MIPI command mode for image mode.<br>1: MIPI video mode for image mode.<br><br>1 = Must fill 1; 0 = Must fill 0; X = Don't care. |    |    |    |    |    |    |    |    |     |
| Restriction |                                                                                                                                            |    |    |    |    |    |    |    |    |     |

## Analog Settings: Analog Settings (90h)

| 90h          | Analog Setting                                    |                    |                   |                  |    |                     |                    |    |    |     |
|--------------|---------------------------------------------------|--------------------|-------------------|------------------|----|---------------------|--------------------|----|----|-----|
|              | W/R                                               | B7                 | B6                | B5               | B4 | B3                  | B2                 | B1 | B0 | Hex |
| Command      | —                                                 | 1                  | 0                 | 0                | 1  | 0                   | 0                  | 0  | 0  | 90h |
| Parameter 1  | W/R                                               | DV_VREFP_OPT [3:0] |                   |                  |    | 0                   | DV_TRIM_DVDD [2:0] |    |    | 70h |
| Parameter 2  | W/R                                               | 0                  | DV_TRIM_OSC [6:0] |                  |    |                     |                    |    |    | 40h |
| Parameter 3  | W/R                                               | 0                  |                   | DV_BG_TRIM [5:0] |    |                     |                    |    |    | 00h |
| Parameter 4  | W/R                                               | 0                  |                   |                  |    | DV_BG_TRIM_TC [3:0] |                    |    |    | 00h |
| Descriptions | DV_VREFP_OPT [3:0]: VREFP Voltage Output Set.     |                    |                   |                  |    |                     |                    |    |    |     |
|              | DV_TRIM_DVDD [2:0]: Trim DVDD Voltage.            |                    |                   |                  |    |                     |                    |    |    |     |
|              | DV_TRIM_OSC [6:0]: Trim OSC Frequency.            |                    |                   |                  |    |                     |                    |    |    |     |
|              | DV_BG_TRIM [5:0]: Trim BG Voltage.                |                    |                   |                  |    |                     |                    |    |    |     |
|              | 1 = Must fill 1; 0 = Must fill 0; X = Don't care. |                    |                   |                  |    |                     |                    |    |    |     |
| Restrictions |                                                   |                    |                   |                  |    |                     |                    |    |    |     |

## IDAC Control: IDAC Control (95h)

| 95h          | IDAC Control                                                                             |             |    |    |    |    |    |    |    |     |
|--------------|------------------------------------------------------------------------------------------|-------------|----|----|----|----|----|----|----|-----|
|              | W/R                                                                                      | B7          | B6 | B5 | B4 | B3 | B2 | B1 | B0 | Hex |
| Command      | —                                                                                        | 1           | 0  | 0  | 1  | 0  | 1  | 0  | 1  | 95h |
| Parameter 1  | W/R                                                                                      | DV_SR [4:0] |    |    |    |    | 0  |    | 1  | 81h |
| Descriptions | DV_SR [4:0]: Trim IDAC current.<br><br>1 = Must fill 1; 0 = Must fill 0; X = Don't care. |             |    |    |    |    |    |    |    |     |
| Restrictions |                                                                                          |             |    |    |    |    |    |    |    |     |

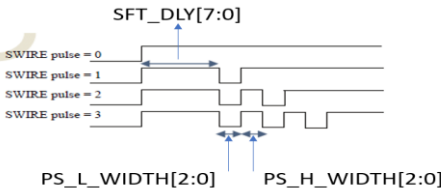
**AVEE Control: AVEE Control(96h)**

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
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| 96h         | AVEE Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                     |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | W / R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | B7                  | B6 | B5 | B4 | B3 | B2 | B1 | B0      | Hex |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
| Command     | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1                   | 0  | 0  | 1  | 0  | 1  | 1  | 0       | 96h |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
| parameter 1 | W / R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | DV_AVEE_OPT[3:0]    |    |    |    | 0  | 0  | 0  | AVEE_ON | 31h |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
| Description | AVEE_ON:<br>1: Enable AVEE according to power sequence.<br>0: Disable AVEE according to power sequence.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                     |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | DV_AVEE_OPT: <table><tr><td>DV_AVEE_OPT</td><td>AVEE Output Voltage</td></tr><tr><td>4'b0000</td><td>-0.8</td></tr><tr><td>4'b0001</td><td>-1.2</td></tr><tr><td>4'b0010</td><td>-1.6</td></tr><tr><td>4'b0011</td><td>-2.0</td></tr><tr><td>4'b0100</td><td>-2.4</td></tr><tr><td>4'b0101</td><td>-2.8</td></tr><tr><td>4'b0110</td><td>-3.2</td></tr><tr><td>4'b0111</td><td>-3.6</td></tr><tr><td>4'b1000</td><td>-4.0</td></tr><tr><td>4'b1001</td><td>-4.4</td></tr><tr><td>4'b1010</td><td>-4.8</td></tr><tr><td>4'b1011</td><td>-5.2</td></tr><tr><td>4'b1100</td><td>-5.6</td></tr><tr><td>4'b1101</td><td>-6.0</td></tr><tr><td>4'b1110</td><td>-6.4</td></tr><tr><td>4'b1111</td><td>-6.8</td></tr></table> |                     |    |    |    |    |    |    |         |     | DV_AVEE_OPT | AVEE Output Voltage | 4'b0000 | -0.8 | 4'b0001 | -1.2 | 4'b0010 | -1.6 | 4'b0011 | -2.0 | 4'b0100 | -2.4 | 4'b0101 | -2.8 | 4'b0110 | -3.2 | 4'b0111 | -3.6 | 4'b1000 | -4.0 | 4'b1001 | -4.4 | 4'b1010 | -4.8 | 4'b1011 | -5.2 | 4'b1100 | -5.6 | 4'b1101 | -6.0 | 4'b1110 | -6.4 | 4'b1111 | -6.8 |
|             | DV_AVEE_OPT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | AVEE Output Voltage |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b0000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -0.8                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b0001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -1.2                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b0010                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -1.6                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b0011                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -2.0                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b0100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -2.4                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b0101                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -2.8                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b0110                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -3.2                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b0111                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -3.6                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b1000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -4.0                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b1001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -4.4                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b1010                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -4.8                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b1011                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -5.2                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b1100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -5.6                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b1101                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -6.0                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b1110                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -6.4                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 4'b1111                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | -6.8                |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
|             | 1 = Must fill 1; 0 = Must fill 0; X = Don't care.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                     |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |
| Restriction |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                     |    |    |    |    |    |    |         |     |             |                     |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |         |      |

**VEMR Control: VEMR Control(97h)**

|             |                                                                                                        |                  |    |    |    |    |    |    |         |     |
|-------------|--------------------------------------------------------------------------------------------------------|------------------|----|----|----|----|----|----|---------|-----|
| 97h         | VEMR Control                                                                                           |                  |    |    |    |    |    |    |         |     |
|             | W / R                                                                                                  | B7               | B6 | B5 | B4 | B3 | B2 | B1 | B0      | Hex |
| Command     | —                                                                                                      | 1                | 0  | 0  | 1  | 0  | 1  | 1  | 1       | 97h |
| parameter 1 | W / R                                                                                                  | DV_VEMR_OPT[3:0] |    |    |    |    |    |    | VEMR_ON | 01h |
| Description | VEMR_ON:<br>1: Enable VEMR according to power sequence.<br>0: Disable VEMR according to power sequence |                  |    |    |    |    |    |    |         |     |
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**SWIRE Control: SWIRE Control (98h)**

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |    |                  |    |                 |        |           |          |     |
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| 98h         | SWIRE Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                 |    |                  |    |                 |        |           |          |     |
|             | W / R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | B7              | B6 | B5               | B4 | B3              | B2     | B1        | B0       | Hex |
| Command     | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1               | 0  | 0                | 1  | 1               | 0      | 0         | 0        | 98h |
| parameter 1 | W / R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0               | 0  | SWIRE_CKSEL[1:0] |    | PS2_EN          | PS1_EN | SWIRE_OPT | SWIRE_ON | 07h |
| parameter 2 | W / R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | PS1_NUM[7:0]    |    |                  |    |                 |        |           |          | 05h |
| parameter 3 | W / R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | PS2_NUM[7:0]    |    |                  |    |                 |        |           |          | 4Fh |
| parameter 4 | W / R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | SFT_DLY[7:0]    |    |                  |    |                 |        |           |          | 04h |
| parameter 5 | W / R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | PS_H_WIDTH[3:0] |    |                  |    | PS_L_WIDTH[3:0] |        |           |          | 22h |
| Description | <div>SWIRE_ON</div> <div>1: Enable swire according to power sequence.</div> <div>0: Disable swire according to power sequence</div> <div>SWIRE_OPT:</div> <div>1: Swire according to power sequence enable on display-on.</div> <div>0: Swire according to power sequence enable on sleep-out.</div> <div>PS1_EN: Enable pulse 1</div> <div>PS2_EN: Enable pulse 2</div> <div>PS1_NUM[7:0]: Set pulse 1 number, max 255.</div> <div>PS2_NUM[7:0]: Set pulse 2 number, max 255.</div> <div>SWIRE_CKSEL:</div> <div>SFT_DLY[7:0]: Swire pulse delay time.</div> <div></div> <div>PS_H_WIDTH: Swire pulse's high time.</div> <div>PS_L_WIDTH: Swire pulse's low time.</div> <div></div> <div>1 = Must fill 1; 0 = Must fill 0; X = Don't care.</div> |                 |    |                  |    |                 |        |           |          |     |
| Restriction |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |    |                  |    |                 |        |           |          |     |

## Image Engine Control (A0h)

| A0h         | Image Engine Control                                                                                                                                                                          |                     |    |    |    |    |    |    |    |     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----|----|----|----|----|----|----|-----|
|             | W / R                                                                                                                                                                                         | B7                  | B6 | B5 | B4 | B3 | B2 | B1 | B0 | Hex |
| Command     | —                                                                                                                                                                                             | 1                   | 0  | 1  | 0  | 0  | 0  | 0  | 0  | A0h |
| parameter 1 | W / R                                                                                                                                                                                         | 0                   |    | 0  | 0  |    | 0  | 0  | 0  | 00h |
| parameter 2 | W / R                                                                                                                                                                                         | INIT_DBV_DSPON[7:0] |    |    |    |    |    |    |    | 80h |
| Description | INIT_DBV_DSPON[7:0]<br>Initial DBV Value on Display ON<br>※DBV register 0x51 is 10-bit, while parameter 2 of 0xA0 (INIT_DBV_DSPON[7:0]) is 8-bit.<br>Only 8 bits can be programmed (written). |                     |    |    |    |    |    |    |    |     |
| Restriction |                                                                                                                                                                                               |                     |    |    |    |    |    |    |    |     |

## Panel Driving Settings1: Panel Driving Settings1 (A1h)

| A1h         | Panel Driving Settings1                                                                     |          |     |    |     |    |           |    |        |     |
|-------------|---------------------------------------------------------------------------------------------|----------|-----|----|-----|----|-----------|----|--------|-----|
|             | W / R                                                                                       | B7       | B6  | B5 | B4  | B3 | B2        | B1 | B0     | Hex |
| Command     | —                                                                                           | 1        | 0   | 1  | 0   | 0  | 0         | 0  | 1      | A1h |
| parameter 1 | W / R                                                                                       | RTN[7:0] |     |    |     |    |           |    |        | 64h |
| parameter 2 | W / R                                                                                       |          |     |    |     |    | RTN[10:8] |    |        | 00h |
| parameter 3 | W / R                                                                                       | VBP[7:0] |     |    |     |    |           |    |        | 08h |
| parameter 4 | W / R                                                                                       | VFP[7:0] |     |    |     |    |           |    |        | 0Ch |
| parameter 5 | W / R                                                                                       | HBP[7:0] |     |    |     |    |           |    |        | 0Ch |
| parameter 6 | W / R                                                                                       | HFP[7:0] |     |    |     |    |           |    |        | 16h |
| Description | RTN: Set internal TE output vsync single on MIPI video mode.                                |          |     |    |     |    |           |    |        |     |
|             | H                                                                                           |          | V   |    | VBP |    | VFP       |    | FR(Hz) |     |
|             | 640                                                                                         |          | 480 |    | 8   |    | 12        |    | 60     |     |
|             | Formula:                                                                                    |          |     |    |     |    |           |    |        |     |
|             | Line time: ((1/ Frame Rate)*1000)/(480 +VBP + VFP)*1000<br>RTN = LineTime / (1 / 3.125MHz ) |          |     |    |     |    |           |    |        |     |
|             | 1 = Must fill 1; 0 = Must fill 0; X = Don't care.                                           |          |     |    |     |    |           |    |        |     |
| Restriction |                                                                                             |          |     |    |     |    |           |    |        |     |

## Panel Driving Settings2: Panel Driving Settings2 (A2h)

| A2h                                               | Panel Driving Settings2                       |                 |           |          |    |    |          |    |    |     |
|---------------------------------------------------|-----------------------------------------------|-----------------|-----------|----------|----|----|----------|----|----|-----|
|                                                   | W / R                                         | B7              | B6        | B5       | B4 | B3 | B2       | B1 | B0 | Hex |
| Command                                           | —                                             | 1               | 0         | 1        | 0  | 0  | 0        | 1  | 0  | A2h |
| parameter 1                                       | W / R                                         | FH              | FV        | CGM[1:0] |    | X  | SPM[2:0] |    |    | 00h |
| parameter 2                                       | W / R                                         | DE_DISPLAY[7:0] |           |          |    |    |          |    |    | 0Ch |
| Description                                       | CGM[1:0]:<br>2'b00: 640x480<br>2'b01: 640x400 |                 |           |          |    |    |          |    |    |     |
|                                                   | SPM[2:0]:                                     |                 |           |          |    |    |          |    |    |     |
|                                                   | SPM[2:0]                                      |                 | Pixel RGB |          |    |    |          |    |    |     |
|                                                   | 3'b000                                        |                 | R         | G        |    | B  |          |    |    |     |
|                                                   | 3'b001                                        |                 | R         | B        |    | G  |          |    |    |     |
|                                                   | 3'b010                                        |                 | G         | R        |    | B  |          |    |    |     |
|                                                   | 3'b011                                        |                 | G         | B        |    | R  |          |    |    |     |
|                                                   | 3'b100                                        |                 | B         | R        |    | G  |          |    |    |     |
|                                                   | 3'b101                                        |                 | B         | G        |    | R  |          |    |    |     |
|                                                   | FV: Select up or down scan directions.        |                 |           |          |    |    |          |    |    |     |
| FH: Select left or right display directions.      |                                               |                 |           |          |    |    |          |    |    |     |
| 1 = Must fill 1; 0 = Must fill 0; X = Don't care. |                                               |                 |           |          |    |    |          |    |    |     |
| Restriction                                       |                                               |                 |           |          |    |    |          |    |    |     |

## Pixel control index: Pixel control index (A8h)

| A8h         | Pixel control index                                                                                  |    |    |    |    |    |    |    |    |     |
|-------------|------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|             | W / R                                                                                                | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 | Hex |
| Command     | —                                                                                                    | 1  | 0  | 1  | 0  | 1  | 0  | 0  | 0  | A8h |
| parameter 1 | INDEX[7:0]                                                                                           |    |    |    |    |    |    |    |    | 00h |
| Description | <p>INDEX[7:0]:<br/>8'h01 = Select TLSB.</p> <p>1 = Must fill 1; 0 = Must fill 0; X = Don't care.</p> |    |    |    |    |    |    |    |    |     |
| Restriction |                                                                                                      |    |    |    |    |    |    |    |    |     |



### Pixel control byte write: Pixel control byte write (A9h)

| A9h         | Pixel control byte write                                                                         |    |    |    |    |    |    |    |    |     |
|-------------|--------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|
|             | W / R                                                                                            | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 | Hex |
| Command     | —                                                                                                | 1  | 0  | 1  | 0  | 1  | 0  | 0  | 1  | A9h |
| parameter 1 |                                                                                                  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | WE | 00h |
| Description | WE:<br>1 = Select AAh byte1 data write.<br><br>1 = Must fill 1; 0 = Must fill 0; X = Don't care. |    |    |    |    |    |    |    |    |     |
| Restriction |                                                                                                  |    |    |    |    |    |    |    |    |     |

### Pixel control data information: Pixel control data information (AAh)

| AAh         | Pixel control data information                                                                                                                                                                                                             |    |    |           |    |    |    |    |    |     |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----------|----|----|----|----|----|-----|
|             | W / R                                                                                                                                                                                                                                      | B7 | B6 | B5        | B4 | B3 | B2 | B1 | B0 | Hex |
| Command     | —                                                                                                                                                                                                                                          | 1  | 0  | 1         | 0  | 1  | 0  | 1  | 0  | AAh |
| parameter 1 |                                                                                                                                                                                                                                            | 0  | 0  | TLSB[5:0] |    |    |    |    |    | 04h |
| Description | TLSB[5:0]:<br>Set TLSB value<br>Frame Rate 60Hz TLSB Max 0x15.<br>Frame Rate 120Hz TLSB Max 0x0A.<br><br>Pixel control code sequence example:<br>A8h 01h.<br>AAh 0Bh.<br>A9h 01h.<br><br>1 = Must fill 1; 0 = Must fill 0; X = Don't care. |    |    |           |    |    |    |    |    |     |
| Restriction |                                                                                                                                                                                                                                            |    |    |           |    |    |    |    |    |     |

## 3. Revision History

| Version No.     | Date      | Page | Description                 |
|-----------------|-----------|------|-----------------------------|
| AN-K0001-240626 | 2024/6/26 | 15   | Application Note Register   |
| AN-K0001-240731 | 2024/7/31 | 16   |                             |
| AN-K0001-250414 | 2025/4/14 | 16   | Add TLSB FR 120 Max setting |